

Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering- Computer Science & Engineering		
Course Name	Environmental Studies	Session	Jan 2026 – June 2026
Course Code	24UNI0106	Semester/Batch	2nd /2025
L-T-P (Per Week)	2-0-0	Course Credits	02
Pre-requisite	NA	NHEQF Level	4.5
Course Coordinator	Dr. Punit Soni	SDG Number	4, 11, 13

1. Objectives of the Course

The objective of this course is to provide knowledge about multidisciplinary nature of environment, various sources of natural energy, ecosystem, social issues and the environment, etc. Students will be evaluated upon achievement in terms of academic:

- To gain knowledge foundational knowledge about key environmental concepts, including ecosystems, biodiversity, natural resource management, and sustainability.
- To develop an understanding of critical environmental challenges, such as climate change, pollution, deforestation, and loss of biodiversity, and their implications for society.
- To learn to propose sustainable solutions to environmental problems by integrating scientific, technological, and policy perspectives.
- To develop and cultivate critical thinking skills to evaluate human impacts on the environment and embrace ethical responsibility in promoting environmental stewardship.
- To analyse local or global initiatives, advocating for environmental conservation and contributing actively to sustainable development goal.

2. Course Learning Outcomes (CLOs)

Student should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Descriptor	No. of Lectures
CLO01	Develop an understanding of ecological systems and processes, equipping students with the ability to analyse environmental impacts in the context of engineering projects.	P1, P2, P4, P6, P9	Q1, Q2	9
CLO02	Integrate principles of sustainable development and environmental management into the design, construction, and operation of engineering solutions.	P1, P2, P5, P6, P8	Q3, Q4	8
CLO03	Acquire skills for environmental policies, laws, and standards relevant to engineering practices, ensuring compliance and promoting responsible development.	P3, P10, P11	Q3, Q4	7
CLO04	Identify and address environmental issues related to industrial processes, waste management, energy use, and resource conservation.	P1, P5, P6, P7	Q1, Q2, Q3	8
Total Contact Hours				32



CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	Type of Assessment
CLO01	M	M	-	L	-	M	-	-	H	-	-	-	-	Summative
CLO02	M	M	-	-	M	H	-	L	-	-	-	-	-	Summative
CLO03	-	-	M	-	-	-	-	-	-	H	H	-	-	Summative
CLO04	L	-	-	-	H	M	M	-	-	-	-	-	-	Summative

L = Low, M = Medium, H = High

3. **Recommended Books:**

- T1: Text Book of Environmental Studies for Undergraduate Courses, Bharucha, Erach, Universities Press (India).
 T2: Environmental Sustainability for Engineers and Applied Scientists – *John Kuczenski & others*, Cambridge University Press.
 T3: Foundation course in Science and Technology “Environment and Resource - Indira Gandhi Open University, New Delhi.

Reference Books:

- R1: An Introduction to Sustainable Development – *Peter Rogers, Kazi Jalal & John Boyd*, Earthscan.
 R2: Environmental Science for Environmental Management – *Tim O’Riordan (Ed.)*, Routledge.

4. **Other readings and relevant websites:**

Sr No	Link of Journals, Magazines, websites and Research Papers
1.	(PDF) ICEMES'2025: International Conference on Energy, Materials & Environmental Science
2.	(PDF) Interrelation of dams sustainability with the local communities and water quality
3.	(PDF) Environmental Science: Advances – four years of diverse and holistic visions

5. **Recommended Tools and Platforms:**

- Travel ESG Reporting Tool
- StaTact
- CODE

6. **Course Plan: Theory****Theory Plan**

Lect. No.	Topic(s)
1-9	Development of the concept of ecosystem, reservoir ecosystem: energy and environment, energy and environment, energy flow in ecosystem, ecological succession,
10-17	Food webs and ecological, forest ecosystem: grassland ecosystem, desert ecosystem, lands a natural resource.
18-25	The changing nature of earth, drought – i, drought – ii, water management practices, fresh water ecology, air pollution: effects and controls of water pollution, soil pollution, effects of noise pollution: nuclear hazards, the price of panacea - biomedical waste, environmental rights: public
26-32	Environmental awareness, Indian legislative steps to protect our environment, global warming: ozone layer depletion, environment and climate change: biodiversity an introduction, biodiversity and its conservation

Lab Plan

-NA-

7. **Delivery/Instructional Resources:**

- Coursera Platform

**Theory Plan:**

Lect. No.	Topics	CLO	Book No, CH No, Page No	TLM	ALM	Web References	Audio-Video
1-9	Development of the concept of ecosystem	CLO1	-	Learner Centered Learning methods	• Video Synthesis • Quiz/Test Questions	https://onlinecourses.swayam2.ac.in/cec25_es01/preview	-
10-17	Food webs and forest ecosystem	CLO2, CLO3	-	Learner Centered Learning methods	• Video Synthesis • Quiz/Test Questions	https://onlinecourses.swayam2.ac.in/cec25_es01/preview	-
18-25	The changing nature of earth	CLO3, CLO4	-	Learner Centered Learning Methods	• Video Synthesis • Quiz/Test Questions	https://onlinecourses.swayam2.ac.in/cec25_es01/preview	-
26-32	Environmental awareness	CLO4	-	Learner Centered Learning Methods	• Video Synthesis • Quiz/Test Questions	https://onlinecourses.swayam2.ac.in/cec25_es01/preview	-

8. Remedial Classes:

-NA-

9. Self-Learning:

Assignments to promote self-learning, survey of contents from multiple sources.

S. No	Topics	CLO	ALM	References/MOOCs
-	-	-	-	-

10. Delivery Details of Content Beyond Syllabus

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S. No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCs
-	-	-	-	-	-

11. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	Type of Component	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Summative	Component 1	Self-Learn Mode and Certificate Submission	01	100%	100	*Submission of required documents, course progress & certificate submission.	CLO01-CLO04
Total			100				

*Note:-Offline/online exam at university- in case of non-availability of course on Coursera.

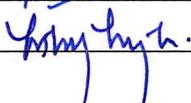
12. Syllabus of the Course:

Subject:			
S. No.	Topic(s)	No. of Lectures	Weightage %
1	Development of the concept of ecosystem, reservoir ecosystem: energy and environment, energy and environment, energy flow in ecosystem, ecological succession,	9	28%
2	Food webs and ecological, forest ecosystem: grassland ecosystem, desert ecosystem, lands a natural resource.	8	25%
3	The changing nature of earth, drought – Drought, water management practices, fresh water ecology, air pollution: effects and controls of water pollution, soil pollution, effects of noise pollution: nuclear hazards, the price of panacea - biomedical waste, environmental rights: public.	7	22%
4	Environmental awareness, Indian legislative steps to protect our environment, global warming: ozone layer depletion, environment and climate change: biodiversity: introduction, biodiversity and its conservation.	8	25%

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Punit Soni	
Program Incharge	Dr. Preetinder Singh Brar	
Pro-Vice Chancellor	Dr. Jaiteg Singh	
Date(DD/MM/YYYY)	27-01-2026	